



Work with ratio and rates

Compare quantities multiplicatively and use rates in meaningful contexts.

Step 45

★ Today I am learning to work with ratio and rates.

1. Understanding Ratio

Write each ratio in simplest form.

- a) $6 : 9 =$ _____
b) $10 : 15 =$ _____
c) $14 : 21 =$ _____
d) $20 : 30 =$ _____

- e) $24 : 36 =$ _____
f) $18 : 27 =$ _____
g) $4 : 16 =$ _____
h) $7 : 28 =$ _____

Write each ratio in the form $1 : n$.

- i) $3 : 9 = 1 :$ _____
j) $2 : 10 = 1 :$ _____
k) $5 : 20 = 1 :$ _____
l) $7 : 21 = 1 :$ _____

Remember:

- Ratio shows the size of one quantity compared to another.
- A ratio $a : b$ means a compared to b .
- Simplify the ratio by dividing both parts by their GCF.

2. Equivalent Ratios

Write two equivalent ratios for each given ratio.

- a) $1 : 2 =$ _____ , _____
b) $2 : 3 =$ _____ , _____
c) $3 : 4 =$ _____ , _____
d) $2 : 5 =$ _____ , _____

- e) $3 : 5 =$ _____ , _____
f) $4 : 7 =$ _____ , _____
g) $5 : 6 =$ _____ , _____
h) $3 : 8 =$ _____ , _____

Remember:

Equivalent ratios are different ratios that are equal in value.

You can find them by multiplying or dividing both parts by the same number.

3. Ratios in Real Life

Solve the problems using ratios.

- a) In a fruit bowl, there are 8 apples and 12 oranges. What is the ratio of apples to oranges?



Answer: _____

- b) A classroom has 15 girls and 10 boys. What is the ratio of girls to boys?



Answer: _____

- c) In a bag, there are 9 red marbles, 6 blue marbles and 3 green marbles. What is the ratio of blue to green marbles?



Answer: _____

- d) A recipe uses 2 cups of flour for every 3 cups of sugar. What is the ratio of flour to sugar?



Answer: _____

4. Rates

Solve the problems using rates.

- a) A car travels 180 km in 3 hours. What is its speed in km per hour?

Answer: _____ km/h

- b) A tap fills a tank with 60 litres of water in 15 minutes. Find the rate in litres per minute.

Answer: _____ L/min

- c) A machine produces 240 bottles in 8 hours. How many bottles are produced per hour?

Answer: _____ bottles/h

- d) If 5 kg of apples cost \$20, what is the cost per kg?

Answer: \$ _____ per kg

- e) A baker makes 120 cupcakes in 3 hours. How many cupcakes does the baker make per hour?

Answer: _____ cupcakes/h

- f) A train travels 450 km in 5 hours. What is its speed in km per hour?

Answer: _____ km/h

Remember:

- Rate is a ratio that compares two different units.
- Common rates:
 - km/h (kilometres per hour)
 - L/min (litres per minute)
 - \$/kg (dollars per kilogram)
- Rate = Total amount ÷ Time or Quantity

5. Compare Quantities Multiplicatively

Solve the problems.

- a) A recipe for lemonade uses 1 part lemon juice to 4 parts water. If you have 2 cups of lemon juice, how many cups of water do you need?

Answer: _____ cups

- b) A map is drawn to a scale of 1 cm represents 5 km. If two cities are 12 cm apart on the map, what is the actual distance between them?

Answer: _____ km

- c) A photo is enlarged using a scale factor of 3. If the original photo is 7 cm wide, how wide is the enlarged photo?

Answer: _____ cm

- d) A bag of rice weighs 2 kg. If you buy 5 of these bags, what is the total weight?

Answer: _____ kg

6. Problem Solving Challenge

Solve the multi-step problems.

- a) A recipe for fruit punch uses apple juice and orange juice in the ratio 2 : 3. If the recipe needs a total of 2 litres of juice, how many litres of each juice are needed?

Answer: Apple juice = _____ L, Orange juice = _____ L

- b) A car uses 7 litres of fuel to travel 84 km.

- a) What is the fuel consumption rate in litres per km?
b) How far can the car travel on 1 litre of fuel?

Answer: a) _____ L/km b) _____ km



My Learning Check

Circle how confident you feel.



I can do it!



Almost there!



I need more help.



This was hard.

Parent Observation (optional)

Please tick the boxes that apply.

- | | |
|--|---|
| ☐ Understands ratio | ☐ Explains thinking clearly |
| ☐ Simplifies ratios correctly | ☐ Works neatly and carefully |
| ☐ Finds equivalent ratios | ☐ Needs more practice |
| ☐ Uses rates in contexts | |
| ☐ Solves multi-step problems | |

Notes: _____

Today I learned...

1 thing I did well:

1 thing I will improve: